

# Mark Zhou

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## EDUCATION

### University of Wisconsin–Madison

*B.S. Computer Sciences, College of Letters & Science*

Madison, WI

*Expected December 2028*

### George School

*High School Diploma, College/University Preparatory Program*

Newtown, PA

*Sep. 2021 – May 2026*

## RESEARCH

### Research Collaborator, Gerych Lab

*Representation Learning & Multimodal ML (Remote); Advisor: Prof. Walter Gerych*

Worcester Polytechnic Institute

*Apr. 2026 – Present*

- Leading implementation and experimental design for [ConceptBasis](#) ([demo](#)), a paper on VLM concept directions
- Built the end-to-end research stack: Gemma-4 labeling, SigLIP2 adapters, and sealed class-disjoint evaluation
- Combined reverse-ridge regression with an orthogonality loss, raising compositional Recall@5 from 19.8% to 74.5%
- Designed five-seed, class-disjoint evaluation across concept prediction, direction overlap, and compositional retrieval

## PROJECTS

### RotateMate – Image Orientation Classifier

*Personal Project*

[github.com/markjough/RotateMate-model](https://github.com/markjough/RotateMate-model)

*2025 – Present*

- Built an end-to-end iOS ML system with custom data, MobileNetV4 training, and SwiftUI deployment
- Quantized a 3M-parameter classifier to 8-bit Core ML and optimized preprocessing for <1 ms on-device inference

### Tiaoqi Master – Chinese Checkers iOS App

*Swift Student Challenge 2026 Winner*

[github.com/markjough/tiaoqi-master](https://github.com/markjough/tiaoqi-master)

*2025 – 2026*

- Built a native Swift 6/SwiftUI learning app with a play-as-you-learn tutorial and adaptive on-device AI
- Implemented time-budgeted iterative-deepening minimax with alpha-beta pruning and heuristic move ordering
- Engineered contextual move commentary and procedural audio synthesis with AVFoundation

### Competitive Programming Library

*Open Source Project*

[github.com/markjough/cp-library](https://github.com/markjough/cp-library)

*2023 – Present*

- Built a CI-verified C++ library spanning data structures, graphs/flow, strings, number theory, and heuristics
- Automated regression tests against Library Checker and Aizu Online Judge through GitHub Actions

## TECHNICAL SKILLS

**ML Systems:** GPU inference profiling, CUDA, vLLM, Core ML, PyTorch, experiment orchestration

**ML Methods:** Vision-language models, contrastive learning, adapter training, multimodal labeling and evaluation

**Languages:** C++, Python, Swift, C, Java, JavaScript, HTML/CSS

**Frameworks & Tools:** AVFoundation, SwiftUI, GitHub Actions, Linux, Git, Xcode

## HONORS & LEADERSHIP

**Competitive Programming:** USACO Platinum; Codeforces Master (2219, top 1%)

**Product Awards:** Apple Swift Student Challenge Winner (2026); c0mpiled Startup School Hackathon II (YC Startup School) – 2nd Place & Most Venture-Backable (\$1,500)

**Contest Wins:** 1st Place – Carnegie Mellon CMIMC Programming (2x), UPenn PClassic Advanced, Massachusetts Computer Science Olympiad, and TJIOI

**Leadership:** Algorithms Club President & Co-Founder; Mathematics Tutor, George School